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Paradise Lost? Estimating the Cost of Hawaii's Pandemic Border Closure with Proximal Inference

Jared Greathouse[†]

[†]*Department of Policy and Management, Georgia State University*
E-mail: `jgreathouse3@student.gsu.edu`

Summary Hawaii's March 2020 mandatory traveler quarantine sealed the state's borders at the same instant the COVID-19 pandemic collapsed global travel demand. For a single treated unit these two forces are perfectly collinear in time, so the conventional synthetic-control toolkit, which matches the treated series to a counterfactual that is itself battered by the pandemic, cannot separate the border-closure *policy* from the coincident *pandemic*. I address this with proximal / negative-control inference in two specifications, Single Proxy Synthetic Control (SPSC) and base Proximal Inference (PI), that *instrument* the latent pandemic factor using insulated employment sectors that carry the pandemic's macro-recession shock yet are immune to the border policy. The two specifications agree, recovering a large, robust tourism-demand collapse (roughly 60–75 percentage points of year-over-year growth for the visitor-volume series); but because the insulated sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly; the estimates from insulated donors alone are therefore best read as *upper bounds in magnitude* on the policy effect and are reported as a robustness band rather than as points. Recovering an actual point requires spanning the missing travel-demand factor from outside the treated economy: adding external travel-demand donors, inbound air travel to Florida and other exposed-but-open leisure destinations that were not locked down, pulls the Visitor Days estimate off its upper bound, and a convergent ensemble of such donors brackets it in a policy-centered band, taking the step from a bound to a band. The paper shows how negative-control inference yields credible causal estimates when a policy and a global shock arrive together.

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1. INTRODUCTION

In 2019, tourism moved $17.75 billion in visitor spending through Hawaii's economy, underwrote $2.07 billion in state tax revenue, and sustained about 216,000 jobs, close to a third of the state's workforce Hawai'i Tourism Authority (2019). In March 2020, Hawaii deliberately switched much of that off. Among all U.S. states it enacted perhaps the most extreme and comprehensive economic shutdown of the COVID-19 pandemic (Yan et al., 2022). Where most regions contracted through voluntary distancing, fear, or uneven mandates, Hawaii chose one of the few deliberate, coordinated state-level shutdowns in the country, a policy aimed not merely at slowing the virus but at suppressing travel and commerce outright. On March 26, 2020, it imposed a mandatory 14-day quarantine on all arrivals, effectively sealing its borders. The outcome was among the starkest in the country: Hawaii went on to record the largest job losses of any U.S. state even as it held among the lowest COVID-19 case rates Bond-Smith and Fuleky (2022). This paper asks what that decision cost.

That question is not merely historical. A governor or health authority weighing whether to seal a jurisdiction (in the next pandemic, or in a climate or geopolitical shock that likewise pits containment against commerce) faces a concrete tradeoff with little credible evidence on the economic side of it (Newman and Noy, 2023). The epidemiological and social effects of COVID-19 interventions have been studied extensively; their economic toll, especially where governments chose maximal containment, far less so. Wuhan's January 2020 lockdown Ke and Hsiao (2022) is among the few rigorous economic estimates of a maximal closure, and work beyond it is rare. Hawaii offers the clearest U.S. instance of the same decision in its starkest form (a whole economy switched off to keep a virus out), and a credible price for it is what a policymaker needs to weigh the closure against the lives it aimed to protect.

Estimating that cost, however, runs into an identification problem that is easy to overlook and fatal if ignored. Hawaii closed its border in the same month the pandemic collapsed global travel demand, and neither force varied before March 2020. For a single treated unit that makes the *policy* and the *pandemic* perfectly collinear in time: every post-treatment month is at once a quarantine month and a pandemic month, so no function of the observed data can separate a response to one from a response to the other. This is not a small-sample nuisance that a larger or cleaner donor pool would cure. It is a property of the data-generating process, and it disables the two research designs one would reach for first.

Difference-in-differences assumes that unit and time effects enter additively: each state's outcome is a fixed state level plus a time shock common to every state, so subtracting untreated states from Hawaii removes the shock and leaves the policy de Chaisemartin and D'Haultfœuille (2022); Roth et al. (2023). That common-shock (parallel-trends) premise is untenable here by construction. Hawaii is the most tourism-dependent state in the country, so the 2020 travel-demand collapse struck it far harder than any candidate control, and a single additive time effect cannot represent a shock whose size scales with how tourism-exposed a state is. Differencing sweeps out a shock that lands equally on treated and controls; it does nothing to one that lands in proportion to tourism exposure, which is exactly the shock at issue.

Synthetic control weakens that premise to a factor model, in which each state's outcome is a few latent, time-varying factors scaled by state-specific loadings, and builds a weighted average of donors whose pre-2020 path tracks Hawaii's. Such a synthetic Hawaii is credible only when the donor weights reproduce Hawaii's factor loadings, so that treated and synthetic respond to every factor alike and the factors, pandemic included, cancel after treatment. The weights are chosen to fit the pre-2020 path, so they can reproduce loadings only on factors that were already moving before 2020. The travel-demand collapse is dormant until March 2020 and then switches on with the policy, leaving no pre-period variation to fit, so a good pre-treatment fit constrains every loading except the one that governs the post-period. The failure is theoretical, not a shortage of data. Even with monthly tourism series for all forty-nine other states, a synthetic Hawaii would remain a no-2020 world rather than a no-quarantine-but-still-pandemic one, because the policy and the pandemic factor are collinear for the treated unit and no reweighting of donors can hold the pandemic on while switching the policy off. Every such counterfactual is at once a no-quarantine and a no-pandemic Hawaii, so the estimate it returns is the border closure plus the travel-demand loss Hawaii would have suffered regardless. Section~2.1 makes this formal: the factor model, the loading-span condition synthetic control requires, and the bias that survives when it fails.

This paper takes that identification problem seriously and resolves it with the right tool. The pandemic confounder is a time-varying latent factor that moves the outcome and coincides with treatment; it cannot be conditioned away because it is unobserved, and it cannot be matched away because it moves any donor too. The *proximal* answer Miao et al. (2018); Tchetgen Tchetgen et al. (2024) is to instrument it: find observable negative controls that are driven by the same pandemic factor but carry no causal path from the treatment, and use them to subtract the pandemic out of the counterfactual. Here those negative controls are insulated employment sectors, hit by the pandemic recession but not downstream of whether tourists can reach Hawaii. I fit two proximal specifications, Single Proxy Synthetic Control (SPSC) Park and Tchetgen Tchetgen (2025) and base Proximal Inference (PI), and report both, so the estimate does not hinge on one estimator's machinery.

Empirically, the two specifications agree: fit on the insulated proxies, SPSC and PI recover the same large tourism-demand collapse, so the result rests on the identification rather than on one estimator. What they identify is a bound, and I treat it as one. The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor, so the tourism estimates absorb both the policy and the demand loss and bound the border-closure effect from above. I report them as a specification band and check each against a relevance diagnostic. Converting that bound into a point requires a proxy that carries the missing travel-demand factor, and I supply one from outside the treated economy: inbound air travel to open, no-lockdown destinations, Florida foremost, which pulls the Visitor Days estimate off its upper bound to a policy-centered point.

A large literature evaluates COVID-19 containment, most of it on the public-health side, where early and aggressive action slows transmission and buys hospital capacity Friedson et al. (2021); Yang (2021); Cerqueti et al. (2021); Bayat et al. (2020). The economic side is thinner and typically studies milder interventions with difference-in-differences or synthetic control Gibson and Sun (2023); Lee and Yang (2022); Ke and Hsiao (2022). Those designs share one requirement, an unaffected comparison unit, and a global pandemic denies it: every U.S. state and comparable island economy took its own pandemic hit, so there is no untreated Hawaii to borrow across space Callaway and Li (2023); Bilgel (2022), while borrowing across time through own-history extrapolation Chen et al. (2024) reconstructs a world with neither the quarantine nor the pandemic and charges the entire 2020 collapse to the policy. The only prior look at Hawaii's pandemic economy is descriptive Bond-Smith and Fuleky (2022). The gap is also a policy gap: as Cerqueti et al. (2021) note, the officials who act first in a crisis are precisely those most exposed to the charge that their intervention looks excessive in hindsight, so a credible price for the most aggressive option is exactly what the record lacks.

The remainder of the paper proceeds as follows. Section~2 states the coincident-shock identification problem and the two proximal estimators. Section~3 describes the data. Section~4 reports the estimates, the specification band, the falsification checks, and the distributional incidence across islands and hotel tiers. Section~5 discusses the policy implications. The standard-error, conformal-band, and base-PI machinery is that of Park and Tchetgen Tchetgen (2025) and Shi et al. (2026), implemented in mlsynth.

2. METHODOLOGY

2.1. The identification problem: a coincident common shock

We know how Hawaii's economy looked before the border closure, and what it did once it mandated the quarantine; but estimating the policy's cost requires a counterfactual, an estimate of how Hawaii's tourism sector would have evolved absent the closure. To build one, we adopt the comparative case study framework, comparing a single treated unit against a set of control units, or donors, that did not enact the policy. Index the N units by j, with $j = 1$ the treated unit (Hawaii) and the other N_0 the control units $\mathcal{N}_0$, and index the T months by t, collected in $\mathcal{T} = \{1, \ldots, T\}$ and split at the final pre-quarantine month T_0 into $\mathcal{T}_1 = \{t \leq T_0\}$ and $\mathcal{T}_2 = \{t > T_0\}$; y_{jt} is unit j's outcome in month t. Write y^1_{1t} and y^0_{1t} for Hawaii's month-t outcome with and without the closure. Only one is ever seen: with $d_t = \mathbf{1}\{t > T_0\}$ the treatment indicator, $y_{1t} = d_t\, y^1_{1t} + (1 - d_t)\, y^0_{1t}$.

Synthetic control builds the counterfactual as a weighted average of the donors. Stacking the treated unit's path into $\mathbf{y}_1 \in \mathbb{R}^T$ and the donors' paths columnwise into $\mathbf{Y}_0 \in \mathbb{R}^{T\times N_0}$, it returns the weight vector $\mathbf{w}$ that minimizes the pre-treatment mean squared error between the treated unit and the weighted donors,

$$\mathbf{w}^* = \underset{\mathbf{w}\in\Delta^{N_0}}{\operatorname{argmin}} \ \left\|\mathbf{y}_1 - \mathbf{Y}_0\mathbf{w}\right\|_2^2. \tag{2.1}$$

The weights are restricted to the probability simplex $\Delta^{N_0} = \{\mathbf{w} \in \mathbb{R}^{N_0}_{\geq 0} : \sum_{j\in\mathcal{N}_0} w_j = 1\}$. Carried past T_0, the fitted values $\mathbf{Y}_0\mathbf{w}^*$ of (2.1) are the counterfactual for the treated unit, and the quarantine's cost is the average treatment effect on the treated over the post-intervention period,

$$\text{ATT} := \frac{1}{n}\sum_{t\in\mathcal{T}_2}\left(y^1_{1t} - y^0_{1t}\right), \qquad n = |\mathcal{T}_2|. \tag{2.2}$$

Whether that estimate is credible depends on the data-generating process. The usual justification for synthetic control is the interactive fixed-effects model Bai (2009); Abadie et al. (2010),

$$y^0_{1t} = \boldsymbol{\mu}_1^\top\boldsymbol{\lambda}_t + e_{1t}, \qquad y_{jt} = \boldsymbol{\mu}_j^\top\boldsymbol{\lambda}_t + e_{jt}, \quad j \in \mathcal{N}_0, \tag{2.3}$$

where $\boldsymbol{\lambda}_t \in \mathbb{R}^r$ collects time-varying factors common to all units, $\boldsymbol{\mu}_j$ collects unit-specific, time-invariant loadings, and $\mathbb{E}[e_{jt} \mid \boldsymbol{\lambda}_t] = 0$ is idiosyncratic error. Under this model, synthetic control is the standard tool, and the theory is precise about when it works. Differencing the treated series against any weighted counterfactual leaves, after some algebra,

$$y_{1t} - \sum_{j\in\mathcal{N}_0} w_j y_{jt} = \underbrace{\left(y^1_{1t} - y^0_{1t}\right)}_{\text{policy effect}} + \underbrace{\left(\boldsymbol{\mu}_1 - \textstyle\sum_{j\in\mathcal{N}_0} w_j\boldsymbol{\mu}_j\right)^\top\boldsymbol{\lambda}_t}_{\text{factor-mismatch bias}} + \left(e_{1t} - \textstyle\sum_{j\in\mathcal{N}_0} w_j e_{jt}\right), \tag{2.4}$$

the policy effect plus a factor-mismatch bias, the residual loading gap $\boldsymbol{\mu}_1 - \sum_j w_j\boldsymbol{\mu}_j$ projected onto the common factors, plus idiosyncratic noise Abadie et al. (2010); Ferman and Pinto (2021); Zhang et al. (2022); Hollingsworth and Wing (2022). The guarantee behind synthetic control is that a long pre-period with a close pre-treatment fit drives that loading gap to zero, forcing the weights to reproduce Hawaii's loadings,

$$\boldsymbol{\mu}_1 = \sum_{j \in \mathcal{N}_0} w_j \, \boldsymbol{\mu}_j. \tag{2.5}$$

For that guarantee to hold however, the same factors and loadings must govern the pre- and post-treatment periods, so that a fit achieved before treatment still holds after it. In the Hawaii case the condition fails. In March 2020 a new coordinate of $\boldsymbol{\lambda}_t$ switches on, the effect of the pandemic, absent from the data before the quarantine and present after, and so perfectly collinear with the treatment over the post-period. The weights are fit on the pre-period, where this factor is silent, so they cannot be chosen to reproduce Hawaii's loading on it: the factor-mismatch term in (2.4) no longer vanishes, and because the factor is unobserved and absent in sample, the bias cannot even be estimated. Practically, this means that even if we had noiseless, observed tourism data across the remaining United States, a good pre-treatment fit built from the donor states no longer certifies the counterfactual, and no reweighting of pre-period donors can subtract a shock the pre-period never saw.

2.2. Proximal inference: single-proxy and base specifications

Fortunately, one solution to this collinearity is the relatively new strand of synthetic control that employs proximal inference Miao et al. (2018); Tchetgen Tchetgen et al. (2024); Shi et al. (2026). The idea is to find a negative control, or proxy: a variable we can use to net the pandemic out of the counterfactual and leave the policy's effect behind. For a proxy to do that, it must be both relevant and exclusionary. A relevant proxy has a strong relationship with the outcome of interest over the pre-policy period, so that it carries a genuine reading of the confounding pandemic factor. An exclusionary proxy is one the policy does not affect, so that reading stays uncontaminated by the treatment we are trying to measure. A variable with both properties moves with the pandemic but not with the border closure, which is exactly what lets the pandemic be subtracted from the counterfactual while the policy effect stays in, pulling apart two forces the collinearity had fused into one.

To state these conditions precisely, collect the candidate proxies at month t in a vector $\mathbf{p}_t$. Relevance asks that some combination of the proxies track the treated unit's untreated outcome,

$$\operatorname{Cov}\big(h^\star(\mathbf{p}_t),\, y_{1t}^0\big) \neq 0 \quad \text{for some } h^\star : \mathbb{R}^{N_0} \to \mathbb{R}, \qquad t \in \mathcal{T}_1, \tag{2.6}$$

the proxy analogue of a strong first stage, whose empirical footprint is the pre-treatment fit; a proxy that does not move with y_{1t}^0 carries no reading of the factor and cannot help. Relevance and exclusion together secure a clean reading of the pandemic, but a reading is not yet a counterfactual. What connects the two is a bridge function: a function $h^\star$ of the proxies that, in expectation, equals the untreated outcome,

$$y_{1t}^0 = \mathbb{E}\big[\, h^\star(\mathbf{p}_t) \,\big|\, y_{1t}^0 \,\big] \quad \text{almost surely}, \qquad t \in \mathcal{T}, \tag{2.7}$$

so that passing the proxies through $h^\star$ returns what Hawaii's tourism would have done absent the closure. In short, a good proxy is one the pandemic drives strongly (relevance) and the policy leaves alone (exclusion), tied to the treated counterfactual through a bridge function. SPSC needs only this single class of proxy, with the treated unit's own history standing in for the second proxy that earlier proximal synthetic control requires Park and Tchetgen Tchetgen (2025); Liu et al. (2024).

Two estimators turn the proxies into a counterfactual, differing only in how they pin down the bridge weights $\mathbf{w}$ in $h^\star(\mathbf{p}_t) = \mathbf{p}_t^\top \mathbf{w}$. Single Proxy Synthetic Control Park and Tchetgen Tchetgen (2025) uses the treated unit's own pre-treatment history as the instrument and fits the bridge by ridge-regularized GMM on the de-trended pre-period series,

$$(\widehat{\boldsymbol{\eta}}, \widehat{\mathbf{w}}_\varrho) = \underset{\boldsymbol{\eta},\mathbf{w}}{\operatorname{argmin}} \left\{ \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta}, \mathbf{w})^\top \, \widehat{\boldsymbol{\Omega}} \, \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta}, \mathbf{w}) + \varrho \, \|\mathbf{w}\|_2^2 \right\}, \tag{2.8}$$

with $\overline{\boldsymbol{\Psi}}$ the averaged moment conditions, $\widehat{\boldsymbol{\Omega}}$ a weight matrix, and the ridge penalty ϱ (cross-validated) stabilizing the weights when the single instrument leaves them under-identified. Base Proximal Inference (PI) is the donors-only estimator of Shi et al. (2026), which in this implementation reduces to a least-squares fit of Hawaii's pre-period series on the donors with a GMM/HAC standard error. Either way the fitted bridge gives the counterfactual $\widehat{y}_{1t}^0 = \mathbf{p}_t^\top \widehat{\mathbf{w}}$ and the ATT (2.2) as the average post-period gap $y_{1t} - \widehat{y}_{1t}^0$; the estimating equations, penalty selection, standard errors, and conformal band are given in Park and Tchetgen Tchetgen (2025) and implemented in mlsynth. Because the treated unit supplies a single instrument against the donors, the analytic standard errors are fragile here, so I lean on the specification band of Section~4 for uncertainty; sharing the identification but not the algorithm, the two estimators' agreement there reflects the proxy design rather than one estimator's machinery.

The proxies fall into two groups, both detailed in Section~3. The first is the $N_0 = 5$ insulated employment sectors: they fell in the pandemic recession, so they carry its macro-recession factor (relevance), but they do not depend on whether visitors can reach Hawaii, so the border closure leaves them untouched (exclusion). The second is a set of external travel-demand donors, inbound air travel to exposed-but-open leisure destinations (Florida foremost), which carry the pandemic's travel-demand-collapse factor while lying outside Hawaii's policy. Both enter the same proxy matrix $\mathbf{P} := \mathbf{Y}_0$, with month-$t$ row $\mathbf{p}_t$; a travel donor widens $\mathbf{P}$ to span a second factor rather than adding a control of a different kind. The split governs what the estimator can recover: the insulated sectors span only the macro-recession factor, so on them alone every tourism estimate is an upper bound in magnitude on the policy effect, absorbing the travel-demand collapse, and adding a travel-demand donor is what turns that bound into a point (Section~4.6). I summarize each fit by its pre-treatment root-mean-squared error $\mathrm{RMSE}_{\mathrm{pre}} = \{T_0^{-1} \sum_{t \in \mathcal{T}_1} (y_{1t} - \widehat{y}_{1t}^0)^2\}^{1/2}$, small relative to the effect when the reconstruction is informative rather than an artifact.

3. DATA

The analysis joins three public sources on calendar month and works throughout in year-over-year (YoY) percentage growth. The YoY transformation removes the strong seasonality of Hawaii tourism and employment, puts series measured in different units on a common scale, and lets each treatment effect be read directly as a percentage-point change in growth. Hawaii's Department of Business, Economic Development & Tourism (DBEDT) supplies the treated tourism and labor series and the insulated employment sectors; the U.S. Bureau of Transportation Statistics (BTS) supplies the external air-travel proxies. The treated panel spans January 1991 to December 2020; the estimation window is described at the end of the section.

3.1. Treated outcomes

The treated series are Hawaii's tourism and labor outcomes, from DBEDT and the Federal Reserve Economic Data (FRED) system: the visitor-volume and price measures, `Visitor Arrivals`, `Visitor Days`, `Occupancy`, `Mean Daily Rate`, and `Revenue per Available Room`, together with `Accommodation Emp` and the broader labor series `Total Leisure Emp`, `Unemp Rate`, and `LFP`. Every one is downstream of the border closure: each moves when visitors' ability to arrive changes, so each can only be a treated outcome, never a control. The five tourism series carry the headline effects, and the labor series are reported alongside them.

3.2. Insulated-sector proxies (the macro negative controls)

The negative-control donors are five insulated employment sectors drawn from the DBEDT seasonally-adjusted Current Employment Statistics, the monthly job count by industry Hawai'i Department of Business, Economic Development and Tourism (2025), expressed in YoY growth: `NatRes_Constr_Emp` (mining, logging and construction), `Wholesale_Emp` (wholesale trade), `Financial_Emp` (financial activities), `HealthCare_Emp` (health care and social assistance), and `Government_Emp`. These industries were hit by the pandemic's macro-recession shock, falling roughly 7–12% in the 2020 trough, but do not depend on whether tourists can physically enter Hawaii, so no causal path runs from the quarantine to them. They are the empirical negative controls of Section~2.2.[1] Figure 1 contrasts the two groups: the treated tourism series collapse by 50–99% at the border closure while the insulated donors dip only modestly, the visual signature of a factor structure in which the donors carry the macro shock and the tourism series carry the macro shock plus the travel-demand collapse plus the policy.

3.3. Open-destination air travel (the travel-demand negative controls)

The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor (Section~2.2), so a second class of proxy is needed to carry that factor: inbound air travel to *exposed-but-open* destinations, places whose air traffic collapsed with the pandemic yet which imposed no traveler quarantine, and so register the travel-demand shock without registering Hawaii's policy. I build these from the BTS Airline On-Time Performance database Bureau of Transportation Statistics (2024), scraped from the agency's airport-statistics portal (https://www.transtats.bts.gov/airports.asp): monthly counts of arriving flights by airport, aggregated to a statewide total for Florida and to the individual airport for the Tampa (TPA), Orlando (MCO), and Phoenix (PHX)

[1] The five are the most insulated supersectors in the CES taxonomy Hawai'i Department of Business, Economic Development and Tourism (2025). I screened the full taxonomy for the negative-control property and deliberately *excluded* the sectors that are downstream of visitor arrivals and would therefore violate it: retail trade (visitor spending), transportation, warehousing and utilities (which bundles air transportation), arts, entertainment and recreation, accommodation and food services, and real estate, rental and leasing (car and vacation rental). Empirically these fell 23–62% in the 2020 trough, tracking the tourism collapse rather than the macro recession, which is exactly the contamination that disqualifies them as controls. A second tier of genuinely insulated sectors, manufacturing, professional and business services, and information, could enlarge the donor pool, but their pre-2020 comovement with the tourism series is no stronger than the five retained donors' ($|\text{corr}| \leq 0.4$), so they do not sharpen identification of the tourism (travel-demand) factor. The one donor that does carry that factor is external, inbound air travel to Florida, added in Section 4.6.

leisure metros, then converted to YoY growth. Florida is the archetype, a large, tourism-dependent state whose government kept its borders open through 2020, and the three metros diversify the proxy across independent leisure markets in states that likewise never locked down. Their identifying role, and why I fit them one at a time rather than en masse, is developed in Section~4.6.

3.4. Panel and estimation window

The treatment indicator `Mandatory Quarantine` switches to 1 in March 2020, giving 10 post-treatment months (March–December 2020). Although the panel ships the full 1991–2020 history, the analysis truncates the pre-period to a recent window beginning January 2014 (74 pre-treatment months). Three decades of stale comovement is less relevant to the 2020 counterfactual than the recent history, and the 2014 window empirically sharpens the pre-treatment fit and yields non-degenerate GMM standard errors that the full-history-plus-detrend combination did not. The full-panel version is available by removing the truncation and enters the robustness band of Section~4.

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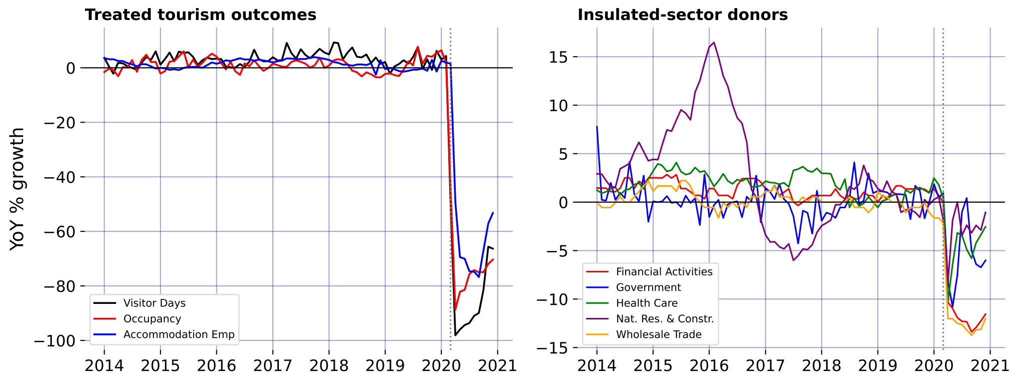

Figure 1: The negative-control premise: treated tourism outcomes collapse while the insulated-sector donors barely dip.

4. RESULTS

4.1. Two proximal specifications, one estimate

Fit on the insulated-sector proxies, the negative controls that carry the pandemic's macro-recession factor but not the border policy, the two headline specifications agree. For Visitor Days, SPSC returns an ATT of -69.3 percentage points and base Proximal Inference (PI) -75.3; across the five headline tourism outcomes the SPSC and PI estimates never differ by more than 10 points. That agreement is the substance of the result, not a coincidence of tuning. SPSC instruments the treated unit's own history under a de-trended ridge-GMM bridge Park and Tchetgen Tchetgen (2025); base PI is the donors-only outcome-bridge estimator of Shi et al. (2026), which in this implementation reduces to a least-squares fit of the treated series on the same proxies. That a regularized instrumental-variable bridge and a plain least-squares fit recover the same effect from the same proxies makes the estimate a property of the proxy design rather than of any one estimator's machinery.

4.2. The pre-treatment fit and the upper bound

Each estimate is only as good as the pre-treatment fit of the bridge, the empirical footprint of the relevance condition (2.6). As Section~2.2 argued, whether that fit reflects genuine relevance depends on which factors the insulated sectors can span. The per-outcome counterfactuals and their conformal uncertainty are shown over time in Figure 4; the readings here summarize them.

Two readings follow. First, the tourism-demand collapse is large and robust: Visitor Days, Visitor Arrivals, Occupancy, and Revenue per Available Room all show a $\sim$ 65–75-point year-over-year drop against the proximal counterfactual, against pre-treatment RMSEs of at most 3.6 growth-rate points across the four (for Visitor Days, 2.6 points against a $\sim$ 70-point effect): the reconstruction is not an artifact of a poor fit. Second, the tourism estimates are nonetheless upper bounds in magnitude on the policy effect, not points: as Section~2.2 shows, the insulated sectors span the macro-recession factor but not the travel-demand-collapse factor, so the tourism loading $\boldsymbol{\mu}_1$ lies outside the donor span (2.5) along the travel-demand direction and the ATT still carries that component. The observed drop is policy plus travel-demand loss. No outcome the insulated sectors span alone delivers a clean point estimate, which is why the headline effects are reported as the band below, and why closing the gap to a point needs the external travel-demand donor of Section~4.6.

Two caveats attach to the GMM standard errors. On this 2014 window they are non-degenerate and usable for the visitor series ($\sim$ 3–8 points), a genuine improvement over the full-history window where de-trending a near-interpolating fit collapses the sandwich toward zero. But two of the labor series remain unreliable, the Unemployment Rate ATT is a relative-growth artifact with a near-zero SE, and the Labor Force Participation SE is tiny, and should not be read at face value. More fundamentally, even a well-behaved SE captures only *sampling* noise around the pre-period bridge, not the *extrapolation* risk of transporting that bridge across the March-2020 structural break, which is the dominant uncertainty here. I therefore do not lean on the SE alone.

4.3. Robustness: a specification band, not a point

Because the honest uncertainty is extrapolation risk rather than sampling noise, I report the headline effects as a band built from specification sensitivity: de-trending on versus off, the B-spline degrees of freedom, the polynomial basis degree, dropping each of three insulated donors individually (the two largest-weight sectors and construction, the proxy most plausibly linked to tourism), and the full-1991 versus 2014 window. Figure 2 collects the minimum, median, and maximum SPSC ATT across these eight specifications for the three best-behaved outcomes.

For the visitor-volume series the band runs roughly from -74 to -59 points (median -69),

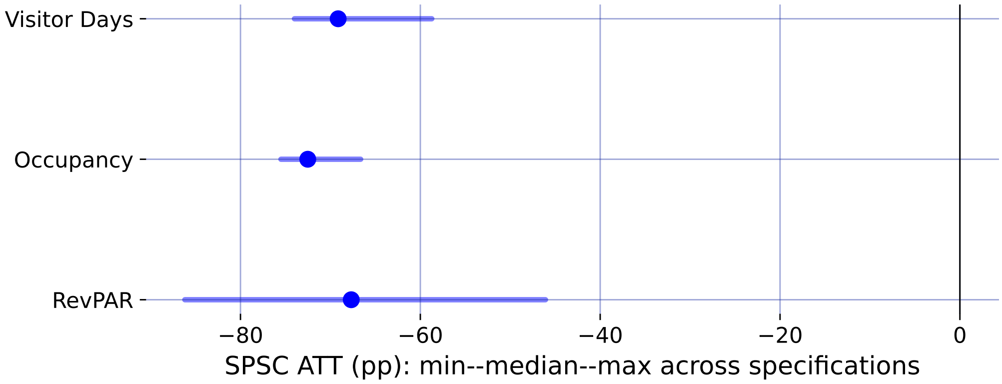

Figure 2: Specification robustness band: median SPSC ATT (point) and min–max range across eight specifications.

so the defensible statement is a $\sim$ 60–75-point *upper-bound* collapse in tourism demand rather than any single figure. Taken together, the diagnostics and the band say the same thing: the border closure sits on top of a large tourism-demand collapse that the insulated sectors cannot fully separate from it, so the tourism numbers bound the policy effect from above. Recovering an actual point estimate, rather than a bound, requires spanning the missing travel-demand factor from outside the treated economy, which is the task of the next section.

4.4. How much would an exclusion violation have to bite?

The identifying content of the insulated proxies rests on exclusion: the border policy must not move them directly (Section~2.2). That is an assumption, and the honest question is how far it can fail before the conclusion changes. Suppose it fails: general-equilibrium spillovers from the closure depress or inflate the insulated sectors, so that over the post-period $p_{jt} = \boldsymbol{\mu}_j^\top \boldsymbol{\lambda}_t + \delta_j d_t + e_{jt}$, with δ_j the direct policy effect on proxy j and $\boldsymbol{\delta} = (\delta_j)_{j \in \mathcal{N}_0}$. Because the bridge weights are estimated over the pre-period, where $d_t = 0$ and the contamination is absent, $\widehat{\mathbf{w}}$ is untouched and the violation reaches the estimate only through the post-period counterfactual:

$$\widehat{\text{ATT}} = \text{ATT} - \boldsymbol{\delta}^\top \widehat{\mathbf{w}}, \tag{4.9}$$

so the bias is linear in the contamination and known up to $\boldsymbol{\delta}$. This makes the sensitivity fully transparent.

For Visitor Days the fitted weights sum to 1.84, so a common spillover of δ points moves the ATT by $1.84\,\delta$. Driving the insulated-only estimate of -69 points to zero would take $\delta \approx +38$ percentage points, the wrong sign (a border closure that *raised* health-care, finance, and government employment growth by nearly forty points) and larger than any insulated sector's *entire* observed 2020 movement, which reached at most 12 points and is itself mostly the macro recession the proxies are meant to carry, not policy spillover. In the economically natural direction, the policy depressing the insulated sectors, $\delta_j < 0$, the bias runs the other way: charging each sector's *full* 2020 decline to the border policy moves the estimate to -83 points, so the reported effect would if anything *understate* the truth. Even an adversarial assignment that signs every δ_j to mask the effect, bounded by each sector's observed movement, shifts the ATT by at most 19 points. Occupancy is more robust still: its weights sum to essentially zero (-0.07), so a common exclusion violation cancels outright. The conclusion does not require the insulated sectors to be perfectly insulated, it survives any exclusion violation of plausible magnitude, and the one plausible direction makes it conservative. The point is sharpest for the most suspect sector: construction tracks hotel building, yet dropping it moves Visitor Days from -69 to -73 points, away from zero rather than toward it, with an unchanged pre-treatment fit (RMSE 2.53 versus 2.59), so the estimate does not lean on the one donor a reader would distrust first.

4.5. A placebo in time

A distinct worry concerns the estimator itself: does the ridge-GMM bridge, with its spline de-trending, manufacture a large effect out of ordinary pre-period variation? The standard falsification is an in-time placebo, assign a fake border closure in a pre-pandemic March, truncate the sample before 2020 so no real shock is in view, and refit. A design that recovers effects only where they exist should return roughly zero.

It does. Across Visitor Days, Occupancy, and Revenue per Available Room, and fake treatment dates in March of 2017, 2018, and 2019, the placebo ATTs never exceed 4 points in magnitude, the scale of the pre-treatment RMSE, and an order of magnitude below the $\sim$ 70-point effects the same procedure returns for 2020. The bridge does not fabricate a collapse out of business-as-usual fluctuation, so the 2020 estimate is a property of the treatment, not of the estimator.

4.6. An ensemble of exposed-but-open destinations

The open-destination air-travel proxies documented in Section~3, inbound flights to Florida and to the Tampa, Orlando, and Phoenix leisure metros, all exposed-but-open, are what close the travel-demand gap that leaves the insulated-only estimates as bounds. Rather than lean on one, I use the whole small ensemble; Florida is the archetype, its air traffic down about 70% at the 2020 trough. Each enters the long panel as its own donor unit (`group = travel-demand`), and, this matters, I fit them *one at a time*, the insulated five plus a single open-destination donor, not all at once (I return to why below).

The proxy must meet the same two conditions as the insulated sectors from the other side: relevance, an exposed-but-open market's inbound air travel loads on the travel-demand factor the insulated sectors cannot span; and exclusion, the donor is a separate economy under no Hawaii quarantine, so it carries the factor without importing the treatment.

The exclusion that does the most work is from Florida's *own* policy, and it is the crux of the design rather than a footnote to it: for the proxy to read the voluntary travel-demand collapse cleanly, Florida's air-travel decline must not be an artifact of a Florida lockdown, and it is not. Florida's mandates were nominal and short-lived. Bollyky et al. (2023) measure state policy stringency by the share of pandemic days each mandate (business and school closures, mask and stay-at-home orders, gathering limits) was in force and place Florida at the permissive end, with fewer restrictions on businesses and individuals than most states Liu et al. (2024); its response was politically driven and centralized to the point of sidelining local emergency-management authority, reopening early and later barring its own localities from reimposing restrictions Witkowski et al. (2023). Behavior matched: at the March 2020 peak Florida's beaches drew spring-break crowds in the thousands against federal guidance, phone-mobility data tracked them dispersing across the eastern United States Business Insider (2020), genomic surveillance later traced variant introductions to that travel Napolitano et al. (2024), and in Pinellas County, part of the Tampa Bay market that furnishes one of the donors here, ethnographers recorded beach tourism continuing with its protocols widely unobserved Porter and McIlvaine-Newsad (2022). That inbound flights collapsed anyway, with little binding policy to explain it, is the point: the decline is the voluntary travel-demand factor the proxy must carry. Puerto Rico, by contrast, imposed a genuine islandwide stay-at-home order in mid-March 2020 and fails the same test, so it is set aside.

The remaining objection, that a proxy like Florida brings idiosyncratic confounders of its own, its hurricane season, drive-to share, is contained on two fronts: the bridge weight is fit on the clean pre-period (a Florida-specific shock that does not track Hawaii before 2020 is down-weighted, and the pre-2020 correlation of +0.35 is what licenses the weight), and the point does not rest on Florida alone, since four independent markets across two states converge on the same band, so a confounder would have to be common to all four and aligned with the March 2020 quarantine, which is the travel-demand collapse the donors are meant to carry.

Figure 3 reports the convergence, and the headline is robustness through agreement. For Visitor Days, every one of the four independent open-destination donors pulls the estimate off the insulated-only bound of -69.3 points into a policy-centered band of roughly -51 to -65 points, and each one *improves* the pre-treatment fit rather than degrading it. That four proxies chosen independently, different cities, two different states, move the estimate the same way and by a similar amount is far stronger evidence that the observed drop is policy plus a real, separable travel-demand collapse than any single proxy could be. Florida gives the cleanest single fit (its Visitor Days pre-treatment RMSE falls from 2.59 to 1.93, bridge weight $w_{\mathrm{FL}} = 0.15$), consistent with its pre-2020 correlation of +0.35 with Hawaii visitor volume.

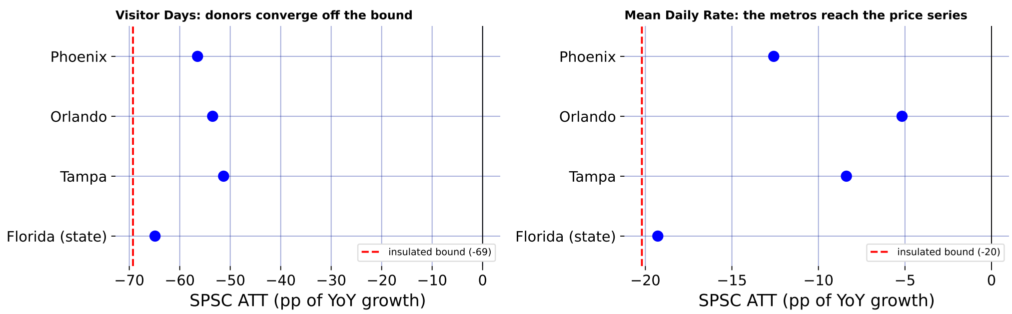

Figure 3: The open-destination ensemble: each donor, fit singly, pulls the estimate off the insulated-only bound.

The ensemble also reaches an outcome a single volume proxy could not. Florida is a flight-*volume* series, so it comoves with Hawaii *visitor volume* but not with hotel prices, and leaves Mean Daily Rate essentially at its insulated-only value of -20.2 points. Phoenix, a desert-resort market whose room rates swing with leisure demand, does comove with Hawaii's daily room rate (pre-2020 correlation +0.36), and adding it pulls Mean Daily Rate from -20.2 toward -12.6 points while *improving* the fit (pre-treatment RMSE 2.37 to 1.81). Different open destinations carry the travel-demand factor as it manifests in different outcomes, volume for the Florida metros, price for Phoenix, so a small, well-chosen ensemble reaches further into the tourism block than any one proxy.

This is the reason for fitting the donors *singly*. Added *en masse*, all four open destinations at once, the near-collinear proxies induce large offsetting ridge weights that *degrade* the fit exactly where a well-chosen single donor helped most, the price and utilization series (for Revenue per Available Room, the pre-treatment RMSE rises from 3.59 insulated-only to 4.39 with the full ensemble), the failure the relevance condition warns of. The lesson favors more *independent* proxies, each read on its own: they corroborate as a convergence band, while collapsing them into a single super-donor is what the collinearity punishes.

Figure 4 is the chapter's summary figure: for every headline outcome it draws the observed series against the two SPSC reconstructions, the insulated-only counterfactual (with its per-period conformal interval as the light spikes), which nets the macro-recession factor and yields the upper bound, and the counterfactual after adding Florida as the anchor donor, which nets the travel-demand factor as well. Base PI's insulated counterfactual is overlaid as a dotted line and sits almost on the SPSC insulated path in every panel, making the two-specification agreement visible rather than asserted. The wedge between the two SPSC lines is where the travel-demand donor does work, visible for the visitor-volume series, invisible for Occupancy, where Florida is (correctly) down-weighted to zero. Reading any single wedge as a clean decomposition would over-interpret it, though: each +donor fit re-solves all bridge weights jointly, so the honest object is the convergence across donors in Figure 3, not one panel here.

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4.7. Who bore the cost: incidence across islands and hotel tiers

The DBEDT workbook that supplies the statewide hotel outcomes also reports them by county and, for revenue per available room, by hotel class, so the same proximal machinery can ask not only how large the tourism collapse was but who absorbed it. For each disaggregated series I refit SPSC on the identical insulated donors and, as in Section~4.6, on the insulated set plus Florida; Figure 5 collects the estimates. Three readings stand out.

First, the demand shock was close to spatially uniform. Hotel occupancy fell between 69 and 75 points across all four counties, so the intuition that the tourism-dependent neighbor islands suffered a deeper demand hit than Oahu is not borne out in occupancy: the rooms emptied everywhere at once.

Second, the collapse ran through volume, not price. The average daily room rate fell only 16 to 23 points, a fraction of the occupancy drop, so hotels largely held their posted rates while rooms went unsold. Because revenue per available room is occupancy times rate, the revenue loss is almost entirely the occupancy loss.

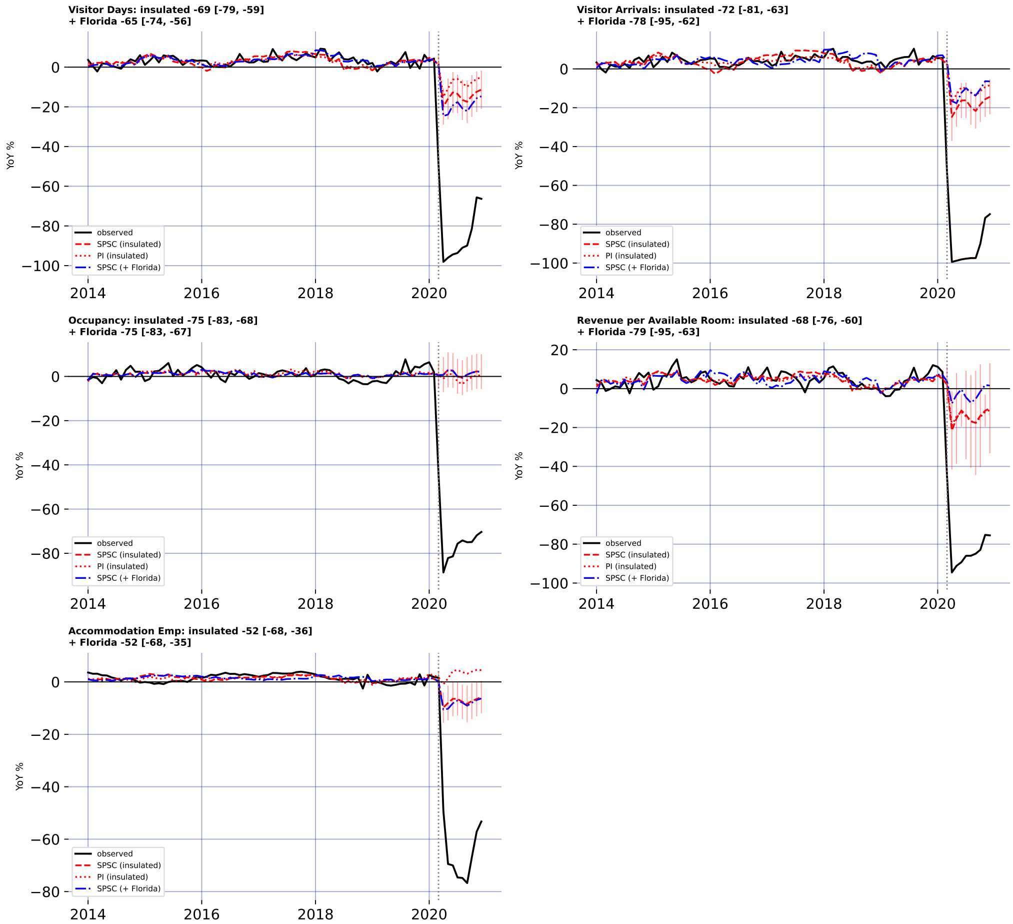

Figure 4: Observed, SPSC and PI insulated counterfactuals, and SPSC + Florida, for five outcomes.

Third, the revenue burden fell regressively across the market. RevPAR by hotel class traces a clear gradient: the luxury and upper-upscale tiers lost the most (89 and 104 points), the upper-midscale tier markedly less among the well-fit segments (58), with midscale and economy least affected of all. Discretionary luxury-leisure travel evaporated where lower-tier and essential lodging demand partly persisted, so the establishments and workers tied to the top of the market bore the sharpest revenue hit.

These disaggregated panels are noisier than the statewide aggregates, and several cells (faded in Figure 5, pre-treatment RMSE above six growth-rate points) carry the near-zero GMM standard errors that Park and Tchetgen Tchetgen (2025) attribute to the rank-deficient weight block when the instrument dimension falls short of the donor count. I read those cells as directional only; the three patterns above rest on the well-fit segments and hold under both the insulated bound and the Florida-adjusted point.

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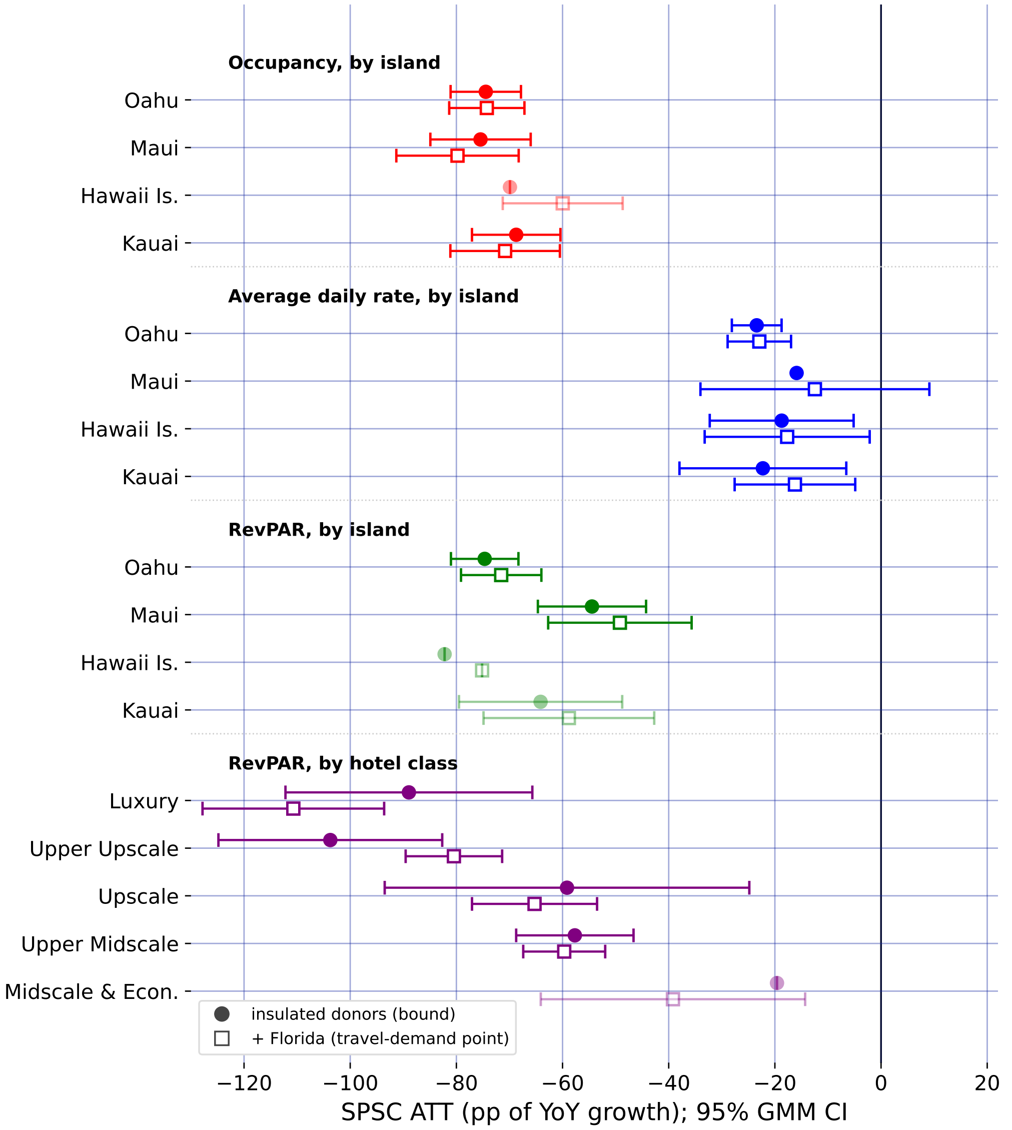

Figure 5: Subgroup ATTs by island and hotel class: insulated bound (circles), Florida-adjusted point (squares).

5. DISCUSSION

The methodological lesson of Hawaii is that a good pre-treatment fit is not identification when a policy and a global shock arrive in the same month. A synthetic control that matches Hawaii to any pre-pandemic counterfactual, whether by borrowing across states or, as own-history extrapolation does, across the treated unit's own past, produces a clean-looking counterfactual and a large ATT, but that ATT is the border-closure effect plus the pandemic's contribution to Hawaii's outcomes, and nothing in the fit reveals the mix. Proximal / negative-control inference is what separates them: instrumenting the latent pandemic factor with insulated employment sectors that carry it but are immune to the policy, both headline specifications, SPSC and base PI, net out the macro-recession component and agree on the result, so the estimate rests on the identification rather than on any single estimator.

At the same time, the analysis is candid about the limit of what the shipped data can identify. The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor: even with an open border, Hawaii tourism would have fallen sharply as people stopped flying. SPSC on insulated sectors alone therefore recovers only the macro component of the counterfactual, and the resulting tourism ATTs over-attribute the collapse to the policy; they are upper bounds in magnitude, which is why I report them as a band rather than as points. Recovering an actual point estimate from insulated donors alone is not possible here, because every treated series loads on the travel-demand factor those donors miss; the point has to be spanned from outside the treated economy, which is what the open-destination donors do. The substantive conclusion is thus deliberately modest and, I would argue, more credible for it: the border closure coincided with a near-total collapse in measured tourism demand, of which a bounded share is causally the policy, with a convergent ensemble of external travel-demand donors pulling the Visitor Days estimate off its upper bound toward a policy-centered band.

Section~4.6 takes the decisive step of closing the two-factor gap with external travel-demand negative controls (inbound air travel to Florida and to a set of other exposed-but-open leisure markets that did *not* lock down), and the result is instructive about both the promise and the care the approach demands. For Visitor Days the open-destination donors do what the design promises, and they do it *in agreement*: fit one at a time, all four independently pull the estimate off the insulated-only bound into a policy-centered band, each improving the pre-treatment fit. Because the donors are chosen independently (different cities, two different states), their convergence is far stronger evidence that the missing factor is real and separable than any single proxy could be. A well-chosen ensemble also reaches further than one volume series: Phoenix, whose resort room rates track leisure demand, pulls the daily-room-rate series that Florida (a volume proxy) leaves essentially untouched. But the ensemble does not mechanically dissolve every bound, and the search must stay disciplined: near-collinear open destinations added *en masse* induce large offsetting weights and hurt the fit, so the donors are best read as a corroborating convergence band; and Puerto Rico, tempting as a fellow island, fails as a control because it imposed its own stay-at-home order in mid-March 2020 and is a contaminated, treated-like unit that, if anything, bounds the effect from the other side.

These estimates are one side of a ledger. They price the short-run economic cost of the border closure through the end of 2020; they do not, on their own, value the public-health benefit (lives saved, hospital strain averted) the closure was meant to buy. A bounded cost, though, is exactly the input a cost-benefit calculation needs, and the surrounding literature supplies the other side. Work on early containment finds that acting even days sooner averts a large share of deaths: Bayat et al. (2020) estimate that locking down ten days earlier could have cut New York deaths by more than 80%. Evidence on comparably aggressive closures finds the economic damage sharp but short-lived Ke and Hsiao (2022). Set side by side, the calculus for a decisive closure is a large but transitory economic loss weighed against a benefit measured in lives, and this paper supplies the first credible, bounded estimate of that loss for the U.S. case. The incidence analysis of Section~4.7 adds that the loss fell unevenly: hardest on the upper tiers of the lodging market, and through lost volume rather than lower prices. What the paper also provides is a template for pricing such policies at all: when the intervention and the crisis it responds to are one and the same event in time, no comparison unit is clean, and the analyst's task is to find a variable that sees the crisis but not the policy.

6. CONCLUSION

This paper estimates the short-run economic consequences of Hawaii's March 2020 mandatory quarantine while confronting the identification problem that its predecessors on this topic could not: for a single treated unit, the border-closure policy and the COVID-19 pandemic are perfectly collinear in time, so no synthetic control that merely matches Hawaii to a pre-pandemic counterfactual can separate the two. I resolve this with proximal inference (negative-control estimators) in two specifications (Single Proxy Synthetic Control and base Proximal Inference) that instrument the latent pandemic factor using insulated employment sectors which carry the pandemic's macro-recession shock but are immune to the border policy.

Estimated by both specifications, the insulated-sector proxies identify a large, robust tourism-demand collapse (about 60–75 points of year-over-year growth for the visitor-volume series). But because those sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly: the insulated-only estimates are upper bounds in magnitude on the policy effect, reported as a robustness band rather than as points. Because every treated series loads on the travel-demand factor those donors miss, the trustworthy point comes only once the missing factor is spanned from outside the treated economy, the role of the external open-destination travel-demand donors.

Methodologically, the study demonstrates how negative-control inference yields credible causal estimates when a policy and a global shock coincide, a structure common to pandemic border closures, but also to climate shocks and other systemic crises where a would-be counterfactual unit is itself engulfed by the event. Substantively, it delivers a bounded, honest estimate of the economic price of Hawaii's "Great Isolation with Aloha characteristics' '. Adding a small ensemble of external no-lockdown travel-demand donors (inbound air travel to Florida and to the Tampa, Orlando, and Phoenix leisure markets) takes the step from a bound to a policy-centered band: fit independently, they converge in pulling the Visitor Days effect toward the policy-only magnitude, and one of them (Phoenix) reaches the room-rate series a single volume proxy cannot. Finishing the job for every outcome asks only for more such open-destination proxies, disciplined by relevance, which the design readily accommodates.

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